Homework - 1

1. Let $p(x) = x^3 - 2 \in \mathbb{Q}[x]$ and α denote the companion matrix of $p(x)$ over $\mathbb{Q}$.
Let $\mathbb{Q}[\alpha] = \mathbb{Q} \oplus \mathbb{Q}\cdot\alpha \oplus \mathbb{Q}\cdot\alpha^2$, as discussed in the class.
Prove: $\mathbb{Q}[\alpha]$ is a subring of $M_3(\mathbb{Q})$, which, in fact, is a ~~sub~~ field.

2. Give an explicit ring isomorphism (k-linear)

$$\frac{k[x]}{\langle f(x) \rangle} \xrightarrow{\sim} k[\alpha]$$

for $f(x) \in k[x]$ a non-constant poly & α its companion matrix $/ k$.
Prove: $k[\alpha] \subseteq M_n(k)$ is a field $\Longleftrightarrow$ f is irreducible in $k[x]$, $n = \deg(f)$.

3. If $\mathbb{Q}[2^{1/3}] := \{ a + b\,2^{1/3} + c\,2^{2/3} \mid a, b, c \in \mathbb{Q} \} \subseteq \mathbb{R}$.
Prove $\mathbb{Q}[2^{1/3}]$ is a subring & a subfield, $\mathbb{Q}$-isomorphic to $\mathbb{Q}[\alpha]$, for α as in problem (1).